A Counterexample to the Group-Ring Determinant Conjecture

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

We disprove the unrestricted group-ring Determinant Conjecture. We construct a finitely generated group G and a square matrix over $\mathbb{Z}[G]$ that is invertible over $\mathbb{Q}[G]$ and has Fuglede–Kadison determinant strictly between zero and one. The logarithmic integral defining the determinant is finite. The construction rests on one external input (Theorem 1.3); everything after that input is proved here.

1 The result

1.1 Conventions

Throughout, G is a discrete group. For a commutative ring R, $R[G]$ is the group ring of *finite* sums $c = \sum_g c_g g$ with $c_g \in R$.

- *Identity coefficient.* $[1]c := c_1$, the coefficient of the identity element.
- *Involution.* For $R \subseteq \mathbb{C}$, $c^* := \sum_g \overline{c_g}\, g^{-1}$. For a matrix A over $R[G]$, $(A^*)_{ij} := (A_{ji})^*$.
- *Projection.* A matrix $P \in \mathrm{M}_m(\mathbb{Q}[G])$ is an *orthogonal projection* if $P = P^* = P^2$.
- *Regular representation.* G acts on $\ell^2(G)$ by the unitaries $(\lambda(g)\xi)(h) = \xi(g^{-1}h)$, so $\lambda(g)\delta_h = \delta_{gh}$. Extend λ linearly to $\mathbb{C}[G]$. For $A \in \mathrm{M}_n(\mathbb{C}[G])$, the operator T_A on $\ell^2(G)^n$ is $(T_A\xi)_i = \sum_j \lambda(A_{ij})\xi_j$. Since λ is a unitary representation, $A \mapsto T_A$ is a unital $*$-homomorphism: $T_{AB} = T_A T_B$ and $T_{A^*} = T_A^*$.
- *Unit vectors.* $\varepsilon_i \in \ell^2(G)^n$ has δ_1 in component i and 0 elsewhere. (The source calls these e_i; they are renamed because e is used below for a reduction of P.) Since $\lambda(g)\delta_1 = \delta_g$,

$$\langle T_A\varepsilon_i, \varepsilon_i\rangle = [1]A_{ii}. \tag{1.1}$$

- *Trace.* For a bounded operator T on $\ell^2(G)^n$, $\tau_n(T) := \sum_{i=1}^n \langle T\varepsilon_i, \varepsilon_i\rangle$. This trace is *unnormalized*: $\tau_n(I_n) = n$. For a group-ring matrix we write $\tau_n(A) := \tau_n(T_A) = \sum_i [1]A_{ii}$, by Equation (1.1).

1.2 The conjecture

For $A \in \mathrm{M}_n(\mathbb{Z}[G])$, the probability spectral measure of the positive operator $T_A^* T_A$ is

$$\mu_A(E) = \frac{1}{n}\sum_{i=1}^{n} \langle \mathbf{1}_E(T_A^* T_A)\varepsilon_i, \varepsilon_i\rangle \quad (E \subseteq [0,\infty) \text{ Borel}).$$

The square-matrix instance of the Determinant Conjecture is the assertion

$$\int_{(0,\infty)} \log t \, d\mu_A(t) \geq 0 \qquad \text{for all } G,\ n,\ A \in \mathrm{M}_n(\mathbb{Z}[G]). \tag{1.2}$$

The zero atom is omitted, and in general the integral may be $-\infty$. The usual conjecture also permits rectangular matrices [7, Conjecture 13.2]; a square counterexample already contradicts it.

Write $\mathcal{N}(G)$ for the group von Neumann algebra. If T_A is invertible, then $\|T_A^{-1}\|^{-2} I \leq T_A^* T_A \leq \|T_A\|^2 I$, so μ_A is supported in a compact subinterval of $(0,\infty)$, on which log is continuous. The spectral theorem then gives $\langle \log(T_A^* T_A)\varepsilon_i, \varepsilon_i\rangle = \int \log t \, d\langle \mathbf{1}_{(\cdot)}(T_A^* T_A)\varepsilon_i, \varepsilon_i\rangle$. Summing over i, the determinant convention for invertible T_A is

$$\log \det_{\mathcal{N}(G)}(T_A) := \frac{1}{2}\tau_n\big(\log(T_A^* T_A)\big) = \frac{n}{2}\int_{(0,\infty)} \log t \, d\mu_A(t). \tag{1.3}$$

For the trivial group this is $\log|\det A|$. This determinant was introduced by Fuglede and Kadison [4].

Remark 1.1 (Matching the row-vector convention). The inversion unitary $\delta_g \mapsto \delta_{g^{-1}}$ on each copy of $\ell^2(G)$ turns left multiplication by g into right multiplication by g^{-1}. It therefore identifies T_A with right multiplication by A^* on row vectors. It fixes δ_1, hence every ε_i, and so preserves the spectral measures above. Our convention and the row-vector convention thus state the same conjecture.

1.3 Main theorem and its one external input

Theorem 1.2. *There are a finitely generated discrete group G_D, an integer $n \geq 1$, and $A \in \mathrm{M}_n(\mathbb{Z}[G_D])$ invertible over $\mathbb{Q}[G_D]$ such that*

$$0 < \det_{\mathcal{N}(G_D)}(T_A) < 1.$$

The integral in Equation (1.2) is finite and strictly negative. Thus the unrestricted group-ring Determinant Conjecture has a negative resolution.

A unital ring is *directly finite* if $ab = 1$ implies $ba = 1$. The proof uses the following scalar failure of direct finiteness.

Theorem 1.3 ([8, Theorem 1.1])**.** *There exist a finite field K_2 of characteristic two, a finitely generated group H_2, and finite sums $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1$ and $b_2 a_2 \neq 1$.*

Unverified step. Theorem 1.3 is an external existence statement that is not proved in the source or in this document, and it cannot be checked from either. It asserts a failure of direct finiteness for a group algebra over a finite field, that is, a counterexample to Kaplansky's direct finiteness conjecture. Combined with the sofic determinant bound of Elek and Szabó [3, Theorem 5], it also yields the existence of a non-sofic group. Both are, as far as we know, open problems, so the claim is extraordinary. Everything downstream (Lemma 3.1 through Theorem 1.2) is conditional on it.

Remark 1.4 (Status of the input). Theorem 1.3 is quoted, not proved or checked, in this document. It is a strong statement, and Theorem 1.2 is exactly as reliable as it is. According to the source, Section 6 of the companion specifies the field, the group presentation and the scalar sums by a terminating finite prescription. Only the existence statement is used here, not an executed numerical instance or a group-word decision procedure. The field K_2 is finite but is not asserted to be $\mathbb{F}_2$. There is no torsion-free hypothesis, and the companion construction uses torsion. Characteristic two is essential below: a result only in odd characteristic would not supply the binary premise.

2 Proof at a glance

Key idea. Call a triple $(P, D, (u, v))$ over a group G a *gap datum of sizes* (m, r) if

- $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$;
- D is an odd positive integer with $DP \in \mathrm{M}_m(\mathbb{Z}[G])$;
- the reduction $\bar{P}$ of P modulo two factors as $\bar{P} = uv$, where u is $m \times r$ and v is $r \times m$ over $\mathbb{F}_2[G]$.

The proof has two independent halves.

1. *A general machine* (Sections 4 and 5). Every gap datum yields an integral matrix A of size $r + m$, invertible over $\mathbb{Q}[G]$, with

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P)) \log q,$$

   where $q > 1$ is a power of two with $q \equiv 1 \pmod{D}$. This is negative as soon as $\tau_m(P) > r$.

2. *A gap datum with* $\tau_m(P) > r$ (Section 3). From the input $a_2 b_2 = 1 \neq b_2 a_2$ we build a gap datum of sizes $(d + 1, d)$ over $G_D = (\bigoplus_{h \in H_2} C_3) \rtimes H_2$ with $\tau_{d+1}(P) = d + \theta$, $\theta > 0$.

Steps and dependencies.

- **Lemma 3.1** (uses Theorem 1.3). Replace the field K_2 by $\mathbb{F}_2$ at the cost of $d \times d$ matrices: $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ with $XY = I_d$ and a nonzero idempotent defect $C = I_d - YX$ satisfying $XC = 0$, $CY = 0$.
- **Proposition 3.2** (uses Lemma 3.1). Some coefficient of some entry of C equals 1. Averaging projections over the C_3 coordinates produce a rational projection p of trace $\theta > 0$ that isolates this coefficient modulo two. Then $P = \mathrm{diag}(I_d, p)$ has trace $d + \theta$, while $\bar{P}$ factors through d columns.
- **Lemma 4.1** (self-contained). The factorization of $\bar{P}$ modulo 2 lifts to a factorization of P modulo $q = 2^k$, by a finite geometric series.
- **Lemma 4.2** (uses Lemma 4.1). With integral lifts U, V of the mod-q factors, an explicit block matrix A is integral, and $A = L_1 B R_1$ for block shears L_1, R_1 and a block-diagonal B built from q and P. Hence A is invertible over $\mathbb{Q}[G]$.
- **Lemma 5.1** (self-contained analysis). $\frac{d}{dx}\tau_n(\log(M^*M)) = 2\,\mathrm{Re}\,\tau_n(M'M^{-1})$.
- **Lemma 5.2** (uses Lemma 5.1). Along the path $M(x) = L_x B R_x$ the quantity $\tau(\log(M^*M))$ is constant.
- **Proposition 5.3** (uses Lemmas 4.2 and 5.2). Hence $\tau(\log(T_A^* T_A)) = \tau(\log(B^*B)) = 2(r - \tau_m(P)) \log q$.
- **Theorem 1.2** (uses Propositions 3.2 and 5.3). Insert $r = d$, $\tau_{d+1}(P) = d + \theta$ to get $\det_{\mathcal{N}(G_D)}(T_A) = q^{-\theta}$.

The projection construction, the modular lifting and the trace calculation have distinct roles; none replaces the scalar existence input.

Renamed symbols. Relative to the source, a few symbols that had two meanings are renamed.

- The unit vectors are ε_i, and the regular representation is $\lambda(g)$.
- The coefficient ring $(\mathbb{Z}/q\mathbb{Z})[G]$ is $\mathcal{R}_q$.
- In Lemma 5.1, the bounds are c_0, c_1, the contraction is Λ, and its norm bound is η.
- In Lemma 5.2, the shear blocks are Γ, Δ.
- Generic elements in the trace identities are α, β.

3 From the scalar defect to a projection

This section produces the gap datum. All group-ring sums are finite, and matrix products are written in the order in which they are multiplied.

Lemma 3.1. *Let K_2 be a finite field of characteristic two and H_2 a group. For $d = [K_2 : \mathbb{F}_2]$, there is an injective unital ring homomorphism*

$$\rho : K_2[H_2] \longrightarrow \mathrm{M}_d(\mathbb{F}_2[H_2]).$$

For $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1 \neq b_2 a_2$, their images $X = \rho(a_2)$, $Y = \rho(b_2)$ satisfy $XY = I_d$ and $C := I_d - YX \neq 0$, with

$$C^2 = C, \qquad XC = 0, \qquad CY = 0. \tag{3.1}$$

This lemma is used once, to feed Proposition 3.2.

Proof. Step 1 (the field). Choose an $\mathbb{F}_2$-basis of K_2. For $\kappa \in K_2$ let $m(\kappa) \in \mathrm{M}_d(\mathbb{F}_2)$ be the matrix of the $\mathbb{F}_2$-linear map $t \mapsto \kappa t$. Then m is additive, $m(1) = I_d$ and $m(\kappa\kappa') = m(\kappa)m(\kappa')$. It is injective: if $m(\kappa) = 0$ then $\kappa = \kappa \cdot 1 = 0$.

Step 2 (the group ring). A matrix over $\mathbb{F}_2[H_2]$ is the same thing as a finite formal sum $\sum_h M_h h$ with $M_h \in \mathrm{M}_d(\mathbb{F}_2)$, and scalar matrices commute with group elements. Define $\rho(\sum_h \kappa_h h) = \sum_h m(\kappa_h)h$. It is additive and unital, and

$$\rho\Big(\sum_{g,h} \kappa_g \kappa'_h \, gh\Big) = \sum_{g,h} m(\kappa_g)m(\kappa'_h)\, gh = \Big(\sum_g m(\kappa_g)g\Big)\Big(\sum_h m(\kappa'_h)h\Big),$$

so ρ is multiplicative. If $\rho(\sum_h \kappa_h h) = 0$, then $m(\kappa_h) = 0$ for each h, so each $\kappa_h = 0$ by Step 1. Thus ρ is injective.

Step 3 (the identities). $XY = \rho(a_2 b_2) = I_d$. By injectivity, $YX = \rho(b_2 a_2) \neq \rho(1) = I_d$, so $C \neq 0$. Using $XY = I_d$:

$$\begin{aligned} &(YX)^2 = Y(XY)X = YX, \text{ hence } C^2 = I_d - 2YX + (YX)^2 = C; \\ &XC = X - (XY)X = 0; \\ &CY = Y - Y(XY) = 0. \end{aligned}$$

□

The group G_D. Let H_2 be a group. In $\bigoplus_{h\in H_2} C_3$, let z_h generate the copy of C_3 labelled by h; distinct z_h commute. H_2 acts by left translation of the labels, and

$$G_D = \Big(\bigoplus_{h\in H_2} C_3\Big) \rtimes H_2, \qquad g\, z_h\, g^{-1} = z_{gh} \quad (g, h \in H_2).$$

Since $H_2 \le G_D$, we have $\mathbb{F}_2[H_2] \subseteq \mathbb{F}_2[G_D]$.

Proposition 3.2. *Let $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$, $d \ge 1$, satisfy $XY = I_d \ne YX$. In the group $G_D = (\bigoplus_{h\in H_2} C_3) \rtimes H_2$ above, there is an orthogonal projection $P \in \mathrm{M}_{d+1}(\mathbb{Z}[1/3][G_D])$ and a finite set $S \subseteq H_2 \setminus \{1\}$ such that*

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad \bar{P} = uv,$$

where u has size $(d+1)\times d$ and v has size $d \times (d+1)$ over $\mathbb{F}_2[G_D]$. The integer $D = 3^{1+|S|}$ clears all coefficients of P.

In the language of Section 2, $(P, D, (u, v))$ is a gap datum of sizes $(d+1, d)$ with trace exceeding d. It is used in the proof of Theorem 1.2.

Proof. Plan. Pick one nonzero coefficient of the defect $C = I_d - YX$. Build a rational projection p that annihilates, by sandwiching, every other group element in that entry of C. Then $P = \mathrm{diag}(I_d, p)$, and the factors u, v are X, Y extended by one row and one column manufactured from C and $\bar{p}$. As in the proof of Lemma 3.1, C satisfies Equation (3.1), which only needs $XY = I_d$.

Step 1 (choose a coefficient). Since $C \ne 0$, some entry C_{ij} has a group element h_0 with nonzero coefficient. That coefficient is 1, because the field is $\mathbb{F}_2$. Multiplying on the left by h_0^{-1} gives

$$h_0^{-1} C_{ij} = 1 + \sum_{s\in S} s, \qquad S := \{h_0^{-1} h : h \ne h_0 \text{ in the support of } C_{ij}\}. \tag{3.2}$$

S is a finite subset of $H_2 \setminus \{1\}$, possibly empty.

Step 2 (the projection p). In $\mathbb{Q}[G_D]$ put

$$p_h = \frac{1 + z_h + z_h^2}{3} \quad (h \in H_2), \qquad p = p_1 \prod_{s\in S} (1 - p_s).$$

- $p_h^2 = p_h$: the nine products in $(1+z_h+z_h^2)^2$ give each element of C_3 three times, so $(1+z_h+z_h^2)^2 = 3(1 + z_h + z_h^2)$.
- $p_h^* = p_h$: inversion permutes $\{1, z_h, z_h^2\}$.
- All p_h commute, because all z_h commute.

A product of commuting self-adjoint idempotents is a self-adjoint idempotent, so $p = p^* = p^2$. Each of the $1 + |S|$ factors of p has denominator 3, so $3^{1+|S|} p \in \mathbb{Z}[G_D]$.

Step 3 ($psp = 0$ for $s \in S$). Conjugation by s is a ring automorphism of $\mathbb{Q}[G_D]$ with $s z_h s^{-1} = z_{sh}$, hence $s p_h s^{-1} = p_{sh}$ and

$$sps^{-1} = p_s \prod_{t\in S} (1 - p_{st}).$$

So sps^{-1} has the factor p_s, while p has the factor $1 - p_s$. All factors commute and $(1 - p_s)p_s = p_s - p_s^2 = 0$, so $p\,(sps^{-1}) = 0$. Therefore

$$psp = p\,(sps^{-1})\, s = 0 \qquad (s \in S). \tag{3.3}$$

The argument does not need the label sets $\{1\} \cup S$ and $\{s\} \cup sS$ to be disjoint. Inverse pairs or finite-order elements in S change nothing.

Step 4 (the trace of p). The labels in $\{1\} \cup S$ are pairwise distinct because $1 \notin S$. Expanding

$$p = 3^{-(1+|S|)} (1 + z_1 + z_1^2) \prod_{s \in S} (2 - z_s - z_s^2),$$

a monomial $z_1^a \prod_s z_s^{b_s}$ with exponents in $\{0, 1, 2\}$ is the identity only if all exponents vanish, because distinct coordinates are independent. Hence

$$\tau_1(p) = [1]p = \frac{1}{3} \prod_{s \in S} \frac{2}{3} = \theta > 0. \tag{3.4}$$

For empty S this reads $p = p_1$ and $\theta = 1/3$.

Step 5 (reduction modulo two). Since 3 is odd, there is a ring homomorphism $\mathbb{Z}[1/3] \to \mathbb{F}_2$. Applied coefficientwise it gives a ring homomorphism $c \mapsto \bar{c}$ from $\mathbb{Z}[1/3][G_D]$ to $\mathbb{F}_2[G_D]$ fixing group elements. So $\bar{p}^2 = \bar{p}$, and $\bar{p}s\bar{p} = 0$ for $s \in S$ by Equation (3.3). With Equation (3.2),

$$\bar{p}\, h_0^{-1} C_{ij}\, \bar{p} = \bar{p}\Big(1 + \sum_{s \in S} s\Big)\bar{p} = \bar{p}^2 + \sum_{s \in S} \bar{p}s\bar{p} = \bar{p}.$$

Step 6 (assembling P, u, v). Let f be the row of length d with entry $\bar{p}h_0^{-1}$ at position i and zero elsewhere. Let w be the column of length d with entry $\bar{p}$ at position j and zero elsewhere. Then $fCw = \bar{p}h_0^{-1}C_{ij}\bar{p} = \bar{p}$ by Step 5. Define

$$P = \operatorname{diag}(I_d, p), \qquad u = \begin{pmatrix} X \\ fC \end{pmatrix}, \qquad v = \begin{pmatrix} Y & Cw \end{pmatrix}.$$

The sizes are $(d+1) \times d$ and $d \times (d+1)$. By Equation (3.1) ($XC = 0$, $CY = 0$, $C^2 = C$) and $XY = I_d$,

$$uv = \begin{pmatrix} XY & XCw \\ fCY & fC^2w \end{pmatrix} = \begin{pmatrix} I_d & 0 \\ 0 & fCw \end{pmatrix} = \operatorname{diag}(I_d, \bar{p}) = \bar{P}.$$

By Step 2, $P = P^* = P^2$ and $3^{1+|S|}P$ is integral. By Equation (3.4), $\tau_{d+1}(P) = d + \tau_1(p) = d + \theta$. □

The rational projection P has trace greater than d, yet its binary reduction factors through d columns. It is the rational projection, not its reduction, that carries the extra trace θ.

4 Lifting the factorization and building the integral matrix

From here until the end of Section 5, G is an arbitrary group and $(P, D, (u, v))$ is a gap datum of sizes (m, r) with $m, r \geq 1$ (Section 2).

Reductions. Let $q = 2^k$, $k \geq 1$, and $\mathcal{R}_q := (\mathbb{Z}/q\mathbb{Z})[G]$. Since D is odd, it is a unit modulo q, so there is a ring homomorphism $\mathbb{Z}[1/D] \to \mathbb{Z}/q\mathbb{Z}$. For a matrix Φ over $\mathbb{Z}[1/D][G]$, write Φ_q for its coefficientwise image over $\mathcal{R}_q$ and $\bar{\Phi}$ for its image over $\mathbb{F}_2[G]$. On integral matrices these are the usual reductions, and reducing Φ_q further modulo two gives $\bar{\Phi}$.

Lemma 4.1. *Let G be a group, $m, r \geq 1$, D an odd positive integer, and $P \in \mathrm{M}_m(\mathbb{Q}[G])$ satisfy $P^2 = P$ and $DP \in \mathrm{M}_m(\mathbb{Z}[G])$. Suppose $\bar{P} = uv$ over $\mathbb{F}_2[G]$, where u and v have respective sizes $m \times r$ and $r \times m$. For $q = 2^k$, $k \geq 1$, the reduction $e = P_q$ has a factorization $e = u_q v_q$ of the same sizes over $(\mathbb{Z}/q\mathbb{Z})[G]$.*

Only idempotence of P is used. The lemma feeds Lemma 4.2.

Proof. Plan. Lift u, v arbitrarily. Their product, compressed by e, differs from e by something even, and an element of the form $e - (\text{even})$ is invertible in the corner ring by a finite geometric series.

Step 1. $e^2 = e$, because reduction is a ring homomorphism. Choose arbitrary lifts u_0, v_0 of u, v over $\mathcal{R}_q$ with finite supports. In the corner ring $e\,\mathrm{M}_m(\mathcal{R}_q)e$, whose identity is e, put

$$Z = eu_0v_0e, \qquad W = e - Z.$$

Step 2 ($W^k = 0$). Modulo two, Z becomes $\bar{P}uv\bar{P} = \bar{P}^3 = \bar{P}$, which is also e modulo two. So every coefficient of every entry of W is even. Each coefficient of W^k is a sum of products of k such coefficients, hence divisible by $2^k = q$, so $W^k = 0$ over $\mathcal{R}_q$. This is nilpotence of the even elements of the corner ring; the corner ring itself need not be nilpotent.

Step 3 (corner inverse). Put $Q = e + W + \cdots + W^{k-1}$; for $k = 1$ this is e. Since $We = eW = W$ and $Z = e - W$,

$$QZ = Q(e - W) = (e + W + \cdots + W^{k-1}) - (W + \cdots + W^k) = e - W^k = e.$$

Thus Q is an inverse of Z in the corner ring, not in the whole matrix ring.

Step 4. Set $u_q = Qeu_0$ and $v_q = v_0e$, of sizes $m \times r$ and $r \times m$. Then $u_qv_q = Q\,(eu_0v_0e) = QZ = e$. Only finite sums and products were used, so supports stay finite. □

Choice of q and the target operator. Fix $q = 2^k > 1$ with $q \equiv 1 \pmod{D}$. Such $k \geq 1$ exists because 2 is a unit modulo the odd integer D: take k a positive multiple of its multiplicative order. For $D = 1$ any $k \geq 1$ works. Let u_q, v_q be as in Lemma 4.1, and choose *arbitrary* lifts $U \in \mathrm{M}_{m\times r}(\mathbb{Z}[G])$, $V \in \mathrm{M}_{r\times m}(\mathbb{Z}[G])$ of them. Define

$$J = -q^{-1}P - (I_m - P), \qquad B = \mathrm{diag}(qI_r, J).$$

Motivation. B expands r coordinate summands by q and contracts the range of P by q^{-1}, so its log-determinant should be $(r - \tau_m(P)) \log q$. But B is not integral. Prescribe upper-left block qI_r and off-diagonal blocks V, U. Block elimination subtracts $U(qI_r)^{-1}V = UV/q$ from the lower-right block. For the Schur complement to be J, the lower-right block must be $UV/q + J$. The signs in J are chosen so that this is $q^{-1}(UV - P) - (I_m - P)$, and $UV \equiv P \pmod{q}$ cancels the denominator.

Lemma 4.2. *Suppose $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$, an odd $D > 0$ clears its denominators, and $\bar{P} = uv$ factors through $r \geq 1$ columns. Let $q = 2^k > 1$ satisfy $q \equiv 1 \pmod{D}$. Choose arbitrary integral lifts U, V of the factors supplied by Lemma 4.1. Then*

$$A = \begin{pmatrix} qI_r & V \\ U & q^{-1}(UV - P - q(I_m - P)) \end{pmatrix} \in \mathrm{M}_{r+m}(\mathbb{Z}[G]) \tag{4.1}$$

has an inverse in $\mathrm{M}_{r+m}(\mathbb{Q}[G])$. Moreover, for $x \in [0, 1]$ and

$$L_x = \begin{pmatrix} I_r & 0 \\ xU/q & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & xV/q \\ 0 & I_m \end{pmatrix},$$

one has

$$M(x) := L_xBR_x = \begin{pmatrix} qI_r & xV \\ xU & x^2UV/q + J \end{pmatrix}, \qquad A = M(1). \tag{4.2}$$

The path $M(x)$ from $B = M(0)$ to $A = M(1)$ is what Proposition 5.3 uses.

Proof. Step 1 (integrality). Write $P + q(I_m - P) = qI_m + (1-q)P$. It is integral because $D \mid (q-1)$ and DP is integral. Its reduction modulo q is $0 + 1 \cdot P_q = e$. The reduction of UV is $u_q v_q = e$ by Lemma 4.1. Hence $UV - P - q(I_m - P)$ is an integral matrix all of whose coefficients are divisible by q, and the lower-right block of Equation (4.1) is integral. The other blocks are integral by construction.

Step 2 (B is invertible over $\mathbb{Q}[G]$). Using $P^2 = P$ and $P(I_m - P) = 0$,

$$(q^{-1}P + (I_m - P))(qP + (I_m - P)) = P + (I_m - P) = I_m,$$

and likewise in the other order. So $J^{-1} = -qP - (I_m - P)$ and $B^{-1} = \operatorname{diag}(q^{-1}I_r, J^{-1})$.

Step 3 (the product). Multiplying in order,

$$BR_x = \begin{pmatrix} qI_r & xV \\ 0 & J \end{pmatrix}, \qquad L_x(BR_x) = \begin{pmatrix} qI_r & xV \\ xU & x^2UV/q + J \end{pmatrix}.$$

At $x = 1$ the lower-right block is $UV/q - q^{-1}P - (I_m - P) = q^{-1}(UV - P - q(I_m - P))$, so $M(1) = A$.

Step 4 (inverse). L_x^{-1} and R_x^{-1} are obtained by negating the off-diagonal block. Hence $A^{-1} = R_1^{-1}B^{-1}L_1^{-1} \in \mathrm{M}_{r+m}(\mathbb{Q}[G])$. More generally $M(x)^{-1} = R_x^{-1}B^{-1}L_x^{-1}$ has entries in $\mathbb{R}[G]$ and depends polynomially on x. □

No identities $PU = U$, $VP = V$ or $V = U^*$ are required of the lifts. The endpoint A is integral; the intermediate $M(x)$ need not be.

Example 4.3 (Sanity check with the trivial group). Let $G = 1$, $m = r = 1$, $D = 1$, $q = 2$.

- $P = (1)$, $u = v = (1)$. Lemma 4.1 gives $u_q = v_q = 1$ in $\mathbb{Z}/2\mathbb{Z}$. With lifts $U = 3$, $V = 1$: $A = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$, $|\det A| = 1 = q^{r - \tau_1(P)} = 2^0$.
- $P = (0)$, $u = v = (0)$. With lifts $U = 2$, $V = 4$: $A = \begin{pmatrix} 2 & 4 \\ 2 & 3 \end{pmatrix}$, $|\det A| = 2 = 2^{1-0}$.

Both agree with Proposition 5.3 below. In both, $\tau_m(P) \le r$; Section 3 exists to produce a group where $\tau_m(P) > r$.

5 The trace calculation

The algebra and its trace. The map $\alpha \mapsto T_\alpha$ on $\mathrm{M}_n(\mathbb{C}[G])$ is injective, since the entries of α can be read off from the vectors $T_\alpha \varepsilon_j$. We therefore write α for T_α. Let $\mathcal{A}_n$ be the operator-norm closure of $\mathrm{M}_n(\mathbb{C}[G])$ in the bounded operators on $\ell^2(G)^n$; it is a unital C^*-algebra. The trace $\tau_n(T) = \sum_i \langle T\varepsilon_i, \varepsilon_i \rangle$ has these properties on $\mathcal{A}_n$.

(T1) τ_n is linear and $|\tau_n(T)| \le n\|T\|$, since each $|\langle T\varepsilon_i, \varepsilon_i\rangle| \le \|T\|$.

(T2) $\tau_n(T^*) = \overline{\tau_n(T)}$.

(T3) $\tau_n(TT') = \tau_n(T'T)$. For $\alpha, \beta \in \mathrm{M}_n(\mathbb{C}[G])$, Equation (1.1) gives

$$\tau_n(\alpha\beta) = \sum_{i,j} [1](\alpha_{ij}\beta_{ji}) = \sum_{i,j} \sum_{g \in G} (\alpha_{ij})_g (\beta_{ji})_{g^{-1}},$$

which is symmetric under $(\alpha, i, g) \leftrightarrow (\beta, j, g^{-1})$. The general case follows by density, (T1) and continuity of multiplication.

Lemma 5.1. *For a continuously differentiable path $M : [0,1] \to \mathcal{A}_n$ of invertible elements,*

$$\frac{d}{dx}\tau_n\big(\log(M(x)^*M(x))\big) = 2\,\mathrm{Re}\,\tau_n\big(M'(x)M(x)^{-1}\big).$$

This is the tracial logarithmic derivative of [4, Lemma 2], written with the unnormalized trace. It is used in Lemma 5.2.

Proof. Plan. Expand $\log N$, $N = M^*M$, in a norm-convergent power series, differentiate term by term, and use cyclicity (T3) to collapse the derivative of each power. No commutation of N and N' is assumed.

Step 1 (uniform bounds). $N = M^*M$ is continuously differentiable with $N' = M'^*M + M^*M'$. Inversion is continuous on invertible elements, so $x \mapsto M(x)^{-1}$ is continuous on $[0,1]$. Put $c_1 = 1 + \sup_x \|M(x)\|^2$ and $c_0 = \inf_x \|M(x)^{-1}\|^{-2} > 0$. For every vector ξ, $\langle N\xi, \xi\rangle = \|M\xi\|^2$ and $\|\xi\| \le \|M^{-1}\|\,\|M\xi\|$, so

$$c_0 I \le N(x) \le c_1 I \quad \text{for all } x, \qquad 0 < c_0 < c_1.$$

Step 2 (series). Put $\Lambda = I - N/c_1$. Then $0 \le \Lambda \le (1 - c_0/c_1)I$, so $\|\Lambda\| \le \eta := 1 - c_0/c_1 < 1$. For $t \in [c_0, c_1]$ the scalar series $\log t = \log c_1 - \sum_{j\ge1}(1 - t/c_1)^j/j$ converges uniformly, so the continuous functional calculus gives

$$\log N = (\log c_1)I - \sum_{j\ge1} \frac{\Lambda^j}{j},$$

norm-convergent uniformly in x.

Step 3 (termwise differentiation). $\Lambda' = -N'/c_1$ is continuous, and $(\Lambda^j)' = \sum_{i=0}^{j-1} \Lambda^i\Lambda'\Lambda^{j-1-i}$, so

$$\left\| \frac{d}{dx}\frac{\Lambda^j}{j} \right\| \le \sup_{y\in[0,1]} \|\Lambda'(y)\|\,\eta^{j-1}.$$

The series of derivatives therefore converges uniformly, and $\log N$ is continuously differentiable with derivative $-\sum_j (\Lambda^j)'/j$.

Step 4 (trace). By (T3), $\tau_n(\Lambda^i\Lambda'\Lambda^{j-1-i}) = \tau_n(\Lambda^{j-1}\Lambda')$ for each i, so $\tau_n((\Lambda^j)')/j = \tau_n(\Lambda^{j-1}\Lambda')$. Since τ_n is continuous and linear by (T1), and $\sum_{j\ge1}\Lambda^{j-1} = (I-\Lambda)^{-1} = c_1 N^{-1}$,

$$\frac{d}{dx}\tau_n(\log N) = -\sum_{j\ge1}\tau_n(\Lambda^{j-1}\Lambda') = \frac{1}{c_1}\sum_{j\ge1}\tau_n(\Lambda^{j-1}N') = \tau_n(N^{-1}N').$$

Step 5 (real part). Insert $N' = M'^*M + M^*M'$ and $N^{-1} = M^{-1}(M^*)^{-1}$. By (T3),

$$\tau_n(N^{-1}M^*M') = \tau_n(M^{-1}M') = \tau_n(M'M^{-1}).$$

By (T3), $MN^{-1} = (M^*)^{-1}$ and (T2),

$$\tau_n(N^{-1}M'^*M) = \tau_n(MN^{-1}M'^*) = \tau_n\big((M'M^{-1})^*\big) = \overline{\tau_n(M'M^{-1})}.$$

The sum is $2\,\mathrm{Re}\,\tau_n(M'M^{-1})$. □

Lemma 5.2. *Let $r, m \ge 1$ and let $B \in \mathcal{A}_{r+m}$ be invertible. Let Γ, Δ have respective sizes $m \times r$ and $r \times m$ over the operator-norm closure of $\mathbb{C}[G]$. For $x \in [0,1]$, let*

$$L_x = \begin{pmatrix} I_r & 0 \\ x\Gamma & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & x\Delta \\ 0 & I_m \end{pmatrix}.$$

Then $\tau_{r+m}(\log((L_xBR_x)^*(L_xBR_x)))$ is constant in x.

Here B is any invertible element. Proposition 5.3 applies the lemma with the B of Section 4, $\Gamma = U/q$ and $\Delta = V/q$.

Proof. Write $M = L_x B R_x$. It is polynomial in x and invertible, with $M^{-1} = R_x^{-1} B^{-1} L_x^{-1}$, where the shear inverses negate the off-diagonal blocks. So Lemma 5.1 applies. By the product rule, $M' = L_x' B R_x + L_x B R_x'$, hence

$$M' M^{-1} = L_x' L_x^{-1} + (L_x B)(R_x' R_x^{-1})(L_x B)^{-1}.$$

Now

$$L_x' L_x^{-1} = \begin{pmatrix} 0 & 0 \\ \Gamma & 0 \end{pmatrix} \begin{pmatrix} I_r & 0 \\ -x\Gamma & I_m \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \Gamma & 0 \end{pmatrix}, \qquad R_x' R_x^{-1} = \begin{pmatrix} 0 & \Delta \\ 0 & 0 \end{pmatrix}.$$

Both have zero diagonal blocks, so trace zero. By (T3) the conjugate $(L_x B)(R_x' R_x^{-1})(L_x B)^{-1}$ also has trace zero. Thus $\tau_{r+m}(M' M^{-1}) = 0$, and Lemma 5.1 gives derivative zero. □

Proposition 5.3. *Let G be a discrete group, $m, r \geq 1$, and let $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$. Suppose an odd $D > 0$ clears its denominators and $\bar{P} = uv$ with $u \in \mathrm{M}_{m,r}(\mathbb{F}_2[G])$ and $v \in \mathrm{M}_{r,m}(\mathbb{F}_2[G])$. For every $q = 2^k > 1$ congruent to one modulo D, the matrix A of Lemma 4.2 satisfies*

$$\tau_{r+m}(\log(T_A^* T_A)) = 2(r - \tau_m(P)) \log q, \tag{5.1}$$

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P)) \log q. \tag{5.2}$$

It is integral and rationally invertible, and its positive-square spectral measure has support in a compact subinterval of $(0, \infty)$. Consequently

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = \frac{2(r - \tau_m(P)) \log q}{r + m}. \tag{5.3}$$

Proof. Step 1. Integrality and the inverse over $\mathbb{Q}[G]$ are Lemma 4.2.

Step 2 (reduce to B). By Equation (4.2), $A = M(1)$ and $B = M(0)$ for the path $M(x) = L_x B R_x$. B is invertible in $\mathcal{A}_{r+m}$ by Step 2 of Lemma 4.2. Lemma 5.2 with $\Gamma = U/q$, $\Delta = V/q$ gives

$$\tau_{r+m}(\log(A^* A)) = \tau_{r+m}(\log(B^* B)).$$

Step 3 (compute at B). Since $P^* = P$ and q is real, $J^* = J$. Since $P(I_m - P) = 0$, $J^2 = q^{-2} P + (I_m - P)$. So

$$B^* B = \mathrm{diag}(q^2 I_r, \; q^{-2} P + I_m - P).$$

This operator is q^2 on the first r summands, q^{-2} on the range of the orthogonal projection P in the last m summands, and 1 on the orthogonal complement of that range. These three subspaces are mutually orthogonal and span, so $\log(B^* B)$ acts on them by $2 \log q$, $-2 \log q$ and 0:

$$\log(B^* B) = 2 \log q \; \mathrm{diag}(I_r, -P), \qquad \tau_{r+m}(\log(B^* B)) = 2(r - \tau_m(P)) \log q.$$

With Step 2 this is Equation (5.1). Half of it is Equation (5.2), by Equation (1.3).

Step 4 (spectral support). $T_A^{-1} = T_{A^{-1}}$ is bounded, since A^{-1} is a finite group-ring matrix. So

$$\|T_A^{-1}\|^{-2} I \leq T_A^* T_A \leq \|T_A\|^2 I,$$

and μ_A is supported in $[\|T_A^{-1}\|^{-2}, \|T_A\|^2] \subset (0, \infty)$. Equation (1.3) with $n = r + m$ turns Equation (5.1) into Equation (5.3). □

6 Assembly and exact determinant

Proof of Theorem 1.2. Step 1. Take K_2, H_2, a_2, b_2 from Theorem 1.3 and put $d = [K_2 : \mathbb{F}_2] \geq 1$. Lemma 3.1 gives $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ with $XY = I_d \neq YX$.

Step 2. Proposition 3.2 gives the group $G_D = (\bigoplus_{H_2} C_3) \rtimes H_2$, a finite set S, and an orthogonal projection $P \in \mathrm{M}_{d+1}(\mathbb{Z}[1/3][G_D]) \subseteq \mathrm{M}_{d+1}(\mathbb{Q}[G_D])$ with

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad D = 3^{1+|S|}, \qquad \bar{P} = uv,$$

where u, v have sizes $(d+1) \times d$ and $d \times (d+1)$, and the odd integer D clears the coefficients of P.

Step 3. Choose $k \geq 1$ with $q = 2^k \equiv 1 \pmod{D}$. Proposition 5.3, with $r = d$ and $m = d + 1$, produces $A \in \mathrm{M}_{2d+1}(\mathbb{Z}[G_D])$, invertible over $\mathbb{Q}[G_D]$, with

$$\log \det_{\mathcal{N}(G_D)}(T_A) = (d - (d + \theta)) \log q = -\theta \log q.$$

Since $q > 1$ and $\theta > 0$,

$$0 < \det_{\mathcal{N}(G_D)}(T_A) = q^{-\theta} < 1. \tag{6.1}$$

Step 4. With $n = 2d + 1$, Equation (5.3) reads

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = -\frac{2\theta \log q}{2d+1} < 0, \tag{6.2}$$

which is finite by the spectral bounds in Proposition 5.3.

Step 5. A finite generating set of H_2 together with z_1 generates G_D, since $z_h = h z_1 h^{-1}$ for every $h \in H_2$. Thus G_D is finitely generated. □

The target group G_D differs from the companion's group H_2 and contains the cyclic coordinates of order three. The example has a finite negative logarithmic determinant and no zero spectral mass. It is a failure of the asserted lower bound, not of determinant class.

7 Context

Nothing in this section is used in the proof. The statements about the literature are reported from the source.

- Determinant bounds are part of the theory of L^2-invariants [7]. Approximation and closure arguments established such bounds for large classes of groups, including amenable groups [9, Theorem 1.21, with its erratum]. Elek and Szabó proved the determinant bound for sofic groups [3, Theorem 5], using finite graph approximations and the integrality of products of nonzero eigenvalues of positive integral matrices. Applying that theorem to the positive integral matrix A^*A shows that G_D is not sofic. This is an external consequence, not an ingredient in constructing A.
- The lower-bound question is distinct from determinant *approximation*. Kammeyer's counterexample to a finite-quotient limiting formula has determinant 2 over the discrete Heisenberg group, while its principal congruence quotients of levels 3^i, $i \geq 1$, have determinants at most $5^{1/3}$ [5, Theorem 2 and Section 2]. That discrepancy does not violate the lower bound one.

- The projections p_h of Proposition 3.2 are related to the commuting wreath-product projections of Dicks and Schick [2, Section 3] (compare [2, Lemmas 3.1 and 3.4]) and of Lehner, Neuhauser and Woess [6, Section 2]. Here they separate one coefficient of a binary defect; no spectral-distribution theorem is imported from those works.

- Lemma 5.1 is the classical tracial logarithmic derivative [4, Lemma 2]. It is also present in the path-integral determinant viewpoint of de la Harpe and Skandalis [1, Lemma 1 and Proposition 2(d)]. Its proof and its application to block shears are given in full above.

8 Verification checklist

One step is marked unverified: the external input, Theorem 1.3. The load-bearing points, in the order a sceptical reader should check them, are these.

1. **The external input.** Theorem 1.3 is quoted from [8] and is not checked here; see Remark 1.4. Everything else is proved in this document.

2. **Coefficient separation.** $psp = 0$ for $s \in S$, and $[1]p = \theta$ (Proposition 3.2, Steps 3 and 4). Check that the conjugation rule $sp_hs^{-1} = p_{sh}$ matches the left-translation action, and that the labels $\{1\} \cup S$ are distinct.

3. **The binary factorization.** $uv = \bar{P}$ (Proposition 3.2, Steps 5 and 6). It uses $XC = 0$, $CY = 0$, $C^2 = C$ from Lemma 3.1 and $\bar{p}h_0^{-1}C_{ij}\bar{p} = \bar{p}$. The coefficient at h_0 being exactly 1 is where characteristic two with field $\mathbb{F}_2$ is used.

4. **The lift modulo q.** W is even, $W^k = 0$ and $QZ = e$ in the corner ring (Lemma 4.1, Steps 2 and 3).

5. **Integrality of A.** The lower-right block needs both $q \equiv 1 \pmod{D}$ and $UV \equiv P \pmod{q}$ (Lemma 4.2, Step 1).

6. **The shear factorization.** L_xBR_x equals the displayed matrix and $A = M(1)$ (Lemma 4.2, Step 3). Example 4.3 tests this numerically.

7. **The logarithmic derivative.** Termwise differentiation and the use of cyclicity in place of commutativity (Lemma 5.1, Steps 3 to 5); the trace-zero computation (Lemma 5.2).

8. **The value at B and the normalizations.** $\log(B^*B) = 2\log q\ \mathrm{diag}(I_r, -P)$ (Proposition 5.3, Step 3); the factors $\frac{1}{2}$ and $n = r + m$ in Equations (1.3) and (5.3); the final substitution $r = d$, $\tau_{d+1}(P) = d + \theta$ (proof of Theorem 1.2, Steps 3 and 4).

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